

Week 2 — Reading Pack

Curated excerpts from industry white papers, organized by learning objective

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Required reading before Week 3 · ≈35 minutes

Assigned alongside: FDA, “Framework for FDA's Real-World Evidence Program” (Dec 2018) — executive summary and Section II; the Claims Data Field Guide (reference).

Why a reading pack. Data vocabulary in pharma commercial is learned from the documents practitioners write, not from textbooks. Below are verbatim excerpts (quoted, with source) from IQVIA white papers in the course library, each chosen to illustrate one of tonight's learning objectives, plus the two FDA definitions every analyst is expected to know. Read the excerpts, answer the reflection questions, and bring one answer to Week 3. Full PDFs are in the course site's “White papers” folder.

Learning objectives for Week 2 (from the syllabus)

1. Distinguish medical claims, pharmacy claims, EHR, and syndicated panel data.
2. Build a basic patient cohort from synthetic claims using diagnosis and procedure codes.
3. Identify common data-quality and re-identification risks in real-world data (RWD).

Key concepts: claims data 101 (header vs. line, place of service, ICD-10, CPT/HCPCS); Rx panel data, longitudinal vs. cross-sectional, patient capture and projection; syndicated reports (DDD, Xponent-style projections) and when to trust them.

A. Data is the constraint, not the idea (objective 1)

Two excerpts set the stakes: the industry's own leaders say their data is rarely ready for the analytics they want, and the arrival of agentic AI turns data quality from a back-office task into a commercial risk.

The IQVIA survey reveals that while over 80% of organizations now prioritize AI investment in key commercial functions, most still face significant data challenges. Only a small minority report their data is “fully sufficient” to support commercial AI applications, while the majority describe their data as “mostly” or “moderately” sufficient. [...] Respondents cited fragmented systems, inconsistent data quality, and limited access to real-time or longitudinal data as key barriers to strengthening their data infrastructure to keep pace with the demands of enterprise-scale AI.

— IQVIA, “AI in Life Sciences Commercialization” (2025 survey of 107 senior commercial leaders), Barriers to scale — 11% ‘fully sufficient’, 53% ‘mostly sufficient’

For many organizations, access to data is no longer the only bottleneck. The harder question is whether the data can be understood in context. A field name, metric, or KPI may be familiar to an experienced analyst, but an AI agent needs explicit meaning. If a commercial metric such as new-to-brand performance over a trailing time period is represented only by shorthand, the system must be able to connect that shorthand to business meaning, lineage, ownership, quality, and approved usage. [...] AI changes the economics of data quality. Historically, poor-quality data often produced isolated reporting errors. AI can operationalize those errors across hundreds or thousands of decisions simultaneously. The question is no longer whether bad data creates risk. The question is how quickly that risk can scale.

— IQVIA, “AI Alone Will Not Modernize the Life Sciences Enterprise” (Silvis, 2026)

Core customer, product, territory, consent, compliance, and performance data require disciplined governance. Execution layers, including campaign design, content personalization, channel orchestration, and local experimentation, can then move faster because they are anchored to trusted foundations. [...] A current master data record may show the latest state of a customer, but many business questions require understanding how that customer changed over time. Without temporal context, semantic clarity, and end-to-end lineage, agents may have access to information but lack the context required to use it responsibly.

— IQVIA, “AI Alone Will Not Modernize the Life Sciences Enterprise” (Silvis, 2026)

Reflection A. The second excerpt uses “new-to-brand performance over a trailing time period” as its example of a metric an AI agent cannot interpret from shorthand. Using tonight's vocabulary, write the full definition of that metric: which dataset (cross-sectional or longitudinal), which grain, which look-back window, and which two blind spots.

B. Fit-for-purpose data and the data continuum (objectives 1 & 3)

Regulators and HTA bodies have a phrase for the choice you make every time you pick a dataset: fit for purpose. The external-comparator paper explains it, and its ‘data continuum’ table (reproduced on tonight's slide 38) ranks the sources you will meet in this course.

Selecting and qualifying the right data source is one of the most consequential decisions sponsors make in an EC programme — and one of the most common reasons EC studies fail at regulatory or HTA review. Sponsors face a recurring tension: the most accessible datasets are rarely the most fit-for-purpose, and the most fit-for-purpose datasets often require investment in curation, linkage, or chart abstraction long before the SAT reads out. [...] The reliability of an EC study is closely tied to how accurate, complete, and relevant its data sources are. Regulators describe “fit-for-purpose” data that have sufficient quality and relevance for the intended regulatory use. This ensures the data is suitable for regulatory decisions without necessarily requiring data to be close to perfect as it is expected for RCT data.

— IQVIA, “Bridging the Evidence Gap: External Comparator Studies from Design to Reimbursement” (2025), Fit-for-purpose data

EC datasets can be drawn from a range of real-world data sources, such as chart reviews, EHRs, patient registries, and the recent increase of digital health tools. Each source comes with specific benefits and drawbacks, but also may be combined, linked or enriched to meet evidentiary standards. For instance, EHRs can provide comprehensive clinical details but may sometimes be incomplete; registries typically use standardized formats but may not be widely applicable; Patient-Generated Health Data (PGHD) from apps, fitness trackers and wearable devices can improve patient focus but often need additional validation before being used for regulatory purposes.

— IQVIA, “Bridging the Evidence Gap,” Fit-for-purpose data

Fit-for-purpose data first. The comparator cohort is only as strong as the data that defines it. Prioritise sources with clinical depth, longitudinal follow-up, and documented data quality over sources that are merely accessible. Chart review remains the gold standard for data richness; registry data offers scale. [...] Match on what matters. Population matching must address the prognostic factors most likely to confound the treatment effect estimate. Work backwards from the known confounders in your indication — do not rely on propensity scores alone when important [confounders are missing].

— IQVIA, “Bridging the Evidence Gap,” EC study design principles at a glance

Note how the paper classifies administrative (claims) databases: relevance “limited to scope of the primary task” (billing), clinical depth low, cost medium. That is not a criticism — it is the precise description of what a

claims file is for. Tonight's notebook builds a treatment-pattern view, which claims do well; it does not attempt to measure response or survival, which they cannot.

What regulators mean by RWD and RWE

From the assigned FDA framework (2018), the two definitions you should be able to recite:

Real-World Data (RWD) are data relating to patient health status and/or the delivery of health care routinely collected from a variety of sources. [...] Real-World Evidence (RWE) is the clinical evidence about the usage and potential benefits or risks of a medical product derived from analysis of RWD.

— FDA, "Framework for FDA's Real-World Evidence Program" (December 2018), Section I

The framework then sets out three questions FDA will ask of any RWE: whether the RWD are fit for use (relevance and reliability), whether the study design can provide adequate scientific evidence, and whether the study conduct meets regulatory requirements. Commercial analysts rarely submit to FDA, but a brand's VP asks exactly the same three questions before acting on a slide.

Missing data and completeness in HTA submissions

Robust data integrity and methodological rigour are essential for the acceptance of RWE in HTA. First, ensuring data completeness is critical. High levels of data missingness can undermine the validity of comparative analyses and introduce bias. HTDs should systematically report the extent and pattern of missing data and pre-specify handling strategies within study protocols and Statistical Analysis Plans (SAPs).

— IQVIA, "The Evolving Role of Real-World Evidence in Oncology HTA Decision-Making" (2025), Methodological rigour

The National Institute for Health and Care Excellence (NICE) Real-World Evidence Framework (2022) emphasises three core principles: ensuring high-quality and relevant data, maintaining transparency and integrity throughout the study process, and using robust analytical methods to minimise bias and characterise uncertainty.

— IQVIA, "The Evolving Role of Real-World Evidence in Oncology HTA Decision-Making," RWE guidelines

Reflection B. Tonight's cohort lost 19 real DLBCL patients (of 500) to its own rules and admitted one impostor. Using the NICE principles above, write the two sentences you would put in a methods appendix to make that trade transparent — one stating the rule and its parameters, one characterising the uncertainty it introduces.

C. Building the cohort: DLBCL and the CAR-T referral funnel (objective 2)

Why we chose DLBCL for the cohort exercise, in the industry's own words — and an example of claims being used as the sampling frame for a market-research study.

Diffuse large B-cell lymphoma (DLBCL) is the most common NHL subtype, comprising approximately 30%–40% of NHL cases, with incidence rising with age.

— IQVIA, "Self-Driving CAR for Hematologic Malignancies" (2025), Introduction

The referrers and CAR T treaters included in the sample were either hematologists or oncologists. Each stakeholder had between 3-30 years of experience and spent at least 50% of their time in direct patient care. IQVIA leveraged 2023 claims data to define high and low thresholds for referring oncologists and treaters in addition to triangulating referral and treatment volumes. [...] CAR T treaters and support staff were defined as high if they had managed more than 20 LBCL patients with CAR T-cell therapy in the three months prior to participating in IQVIA's survey.

— IQVIA, “The Path to CAR T-Cell Therapy: Barriers and Opportunities” (2025), Sample

The results of the IQVIA commissioned research on the challenges to uptake of CAR T-cell therapies are striking. Two out of three eligible patients (relapsed or refractory LBCL) are never referred to a treatment center and less than two out of 10 patients ultimately receive CAR T-cell therapy treatment. These findings highlight the significant gulf between the clinical potential of CAR T-cell therapy and the real world.

— IQVIA, “The Path to CAR T-Cell Therapy,” Conclusion

Barriers exist for the LBCL patients who are successfully referred and reach an ATC, with approximately half of these patients exiting the treatment journey before they progress to the apheresis/patient prep stage. [...] Nearly one in four patients who reach a CAR T center fall out of the process after the Initial Screening or Formal / Clinical Evaluation stages as the treating physician at the ATC deems them ineligible. [...] It is also possible that patients who were in good health at the time of their referral may have deteriorated by the time they reach an ATC and are no longer eligible. While patients are ideally able to transition efficiently through the CAR T treatment journey, real-world data illustrate that variation in administration processes and capacity among ATCs can extend the waiting period.

— IQVIA, “The Path to CAR T-Cell Therapy,” Barriers experienced at ATCs

Reflection C. The referral funnel above (eligible → referred → reaches center → screened → apheresis → infused) is a patient journey. For each step, say whether a claims file could observe it, and with which code type (ICD-10, CPT, HCPCS/J or Q-code, NPI, place of service) — or whether it is invisible in claims and needs another source.

D. De-identification, linkage and what AI changes (objective 3)

The last excerpts come from IQVIA's own description of how a data vendor protects patient-level data — useful precisely because it names the mechanisms (pseudonymization, deterministic vs. probabilistic linkage, reconstruction risk) that appear in every data license you will sign.

Robust de-identification methods, which remove identifying elements, can be used but the industry lacks widespread adoption of standardized practices. This absence of fixed standards provides space to explore forward-looking approaches, especially in light of emerging AI/ML threats that will need to be addressed.

— IQVIA, “Trustworthy AI/ML for Patient Analytics and Research” (2025), Introduction

At the entry points, each data stream is pseudonymized before ingestion. Pseudonymization replaces sensitive identifiers with unique, random pseudonyms that have no intrinsic meaning and are internal to the federated modeling process. To maximize separation of duties and enhance trust, this function can be handled by a neutral Third-Party Provider (TTP), ensuring that no internal team or downstream process ever has access to unique identifiers. Depending on the design, pseudonymization can support deterministic linkage (the same input always maps to the same output, enabling consistent matching across datasets) or probabilistic linkage (which allows matching with a configurable degree of fuzziness to accommodate natural data variability).

— IQVIA, “Trustworthy AI/ML for Patient Analytics and Research,” Source ingestion: pseudonymization and segregation

Federated learning architectures are designed to minimize risk by decentralizing data. However, when using raw or even de-identified features, modern AI systems could draw connections and surface features that humans might miss. When applied to high-dimensional datasets, especially ones that combine various sources of health information, even relatively sparse information can yield sensitive inferences. This raises the possibility that models may inadvertently disclose sensitive information or infer sensitive health attributes, undermining privacy. Adversarial actors can exploit poorly abstracted model inputs or outputs

through inference, reconstruction, or correlation attacks. So, while traditional de-identification techniques such as pseudonymization, generalization, or suppression reduce identifiability, our concern is mitigating the ability of systems to infer attributes under adversarial conditions.

— IQVIA, “Trustworthy AI/ML for Patient Analytics and Research,” Beyond de-identification: managing reconstruction risk

This approach is a response to today's threat landscape — including risks such as AI model inversion, data reconstruction, linkage attacks, and the misuse of data across distributed systems — and serves as a blueprint for sustainable innovation. [...] Aligned with global standards such as ISO/IEC 42001 AI Management System and frameworks by the U.S. National Institute of Standards and Technology (NIST), this system ensures continuous oversight, minimizes AI/ML risks, and promotes defensible AI.

— IQVIA, “Trustworthy AI/ML for Patient Analytics and Research,” Heightened care in AI/ML

Reflection D. Tonight's exercise asked which fields would re-identify patient 7. The excerpts describe three families of defence: pseudonymization/suppression/generalization, abstraction to group-level trends, and governance/monitoring. For a term project that trains a model on patient-level (synthetic) data, write one concrete step you would take from each family — and one thing your model would lose because of it.

E. Optional: PMSA items for Week 2

- **PMSA Webinar — “Improving Completeness and Accuracy of Real World Data.”** Direct treatment of the data-quality problem in claims and RWD; the practitioner version of tonight's Part 3 checklist.
- **PMSA Webinar — “Combine Advanced Analytics and Real-World Data to Accurately Identify Treatment Drivers.”** A case study on what drives treatment decisions versus what is claimed — a useful contrast to the ‘spend caused TRx’ framing you will meet in Week 3.
- **JPSMA — “Enhancing Patient Classification and Staging in RWD Using Machine Learning” (ZS Associates).** Machine learning on claims for patient classification and staging: the next step after tonight's rule-based cohort.

PMSA member login may be required for some items; the instructor can share slides where licensing permits.